Status	Finished
Started	Monday, 19 October 2026, 9:14 PM
Completed	Monday, 19 October 2026, 9:18 PM
Duration	4 mins 19 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are split again into smaller sublists
b. They are compared element by element
c. They are discarded
d. They are combined to form a single sorted list

The correct answer is: They are combined to form a single sorted list

Question **2**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It complicates the selection of items
b. It makes it easier to select items based on their relationship to the rest of the items
c. It makes the data unsorted
d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Quick sort
- b. Heap sort
- c. Merge sort
- d. Selection sort

The correct answer is:
Merge sort

Question **4**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the smallest element
- b. Bubbling up the largest element to its correct position
- c. Sorting the entire list in one pass
- d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position

Question **5**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Sorting
- b. Searching
- c. Rearranging
- d. Merging

The correct answer is: Sorting

Question **6**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It only works with integer arrays
b. It requires external libraries
c. It is less efficient than custom sorting algorithms
d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **7**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Extraction
b. Selection
c. Distribution
d. Insertion

The correct answer is:
Insertion

Question **8**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It makes the search operation slower
b. It has no effect on the search operation
c. It increases the number of comparisons needed
d. It allows for faster searching

The correct answer is: It allows for faster searching

Question **9**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Linear Search
b. Binary Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **10**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Arranging data in a particular format
b. Inserting data into a list
c. Searching for data in a list
d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **11**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of non-duplicate keys
b. A sorting algorithm is stable if it preserves the order of duplicate keys
c. A sorting algorithm is stable if it preserves the order of all keys
d. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Merge Sort
- b. Bubble Sort
- c. Insertion Sort
- d. Quick Sort

The correct answer is: Merge Sort

Question **13**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is based on the divide-and-conquer approach
- b. It does not require recursion
- c. It sorts data using a single pass
- d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It divides the list into two halves, sorts each half, and then merges them
- b. It repeatedly finds the minimum element and moves it to the sorted part of the list
- c. It compares adjacent elements and swaps them if necessary
- d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **15**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort does not guarantee sorted order
b. Bubble Sort is less efficient for large lists
c. Bubble Sort cannot handle duplicate elements
d. Bubble Sort is difficult to implement

The correct answer is: Bubble Sort is less efficient for large lists